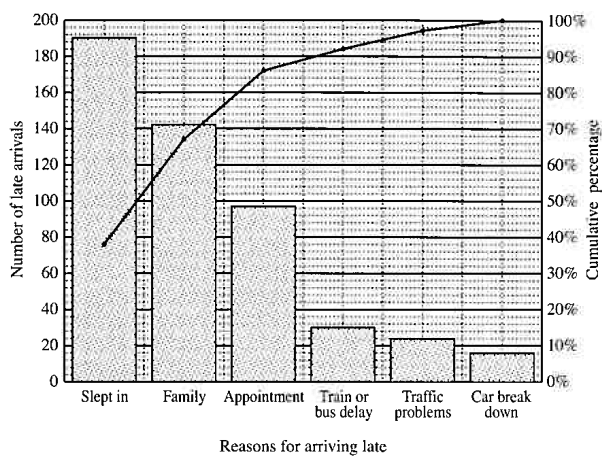


- 1 A school collected data related to the reasons given by students for arriving late. The Pareto chart shows the data collected.



What percentage of students gave the reason 'Train or bus delay'?

- A 6% B 15%
C 30% D 92% (1 mark)

(Q10, 2019 HSC) **Hard**

- 2 A survey asked the following question.
'How many brothers do you have?'

How would the responses be classified?

- A Categorical, ordinal
B Categorical, nominal
C Quantitative, discrete
D Quantitative, continuous (1 mark)

(Q3, 2018 HSC) **Easy**

- 3 A factory's quality control department has tested every 50th item produced for possible defects.

What type of sampling has been used?

- A Random B Stratified
C Systematic D Quantitative (1 mark)

(Q4, 2017 HSC) **Easy**

- 4 Which set of data is classified as categorical and nominal?

- A blue, green, yellow
B small, medium, large
C 5.2 cm, 6 cm, 721 cm
D 4 people, 5 people, 9 people (1 mark)

(Q7, 2016 HSC) **Medium**

- 5 On a school report, a student's record of completing homework is graded using the following codes.

C = consistently
U = usually
S = sometimes
R = rarely
N = never

What type of data is this?

- A Categorical, ordinal
B Categorical, nominal
C Quantitative, continuous
D Quantitative, discrete (1 mark)

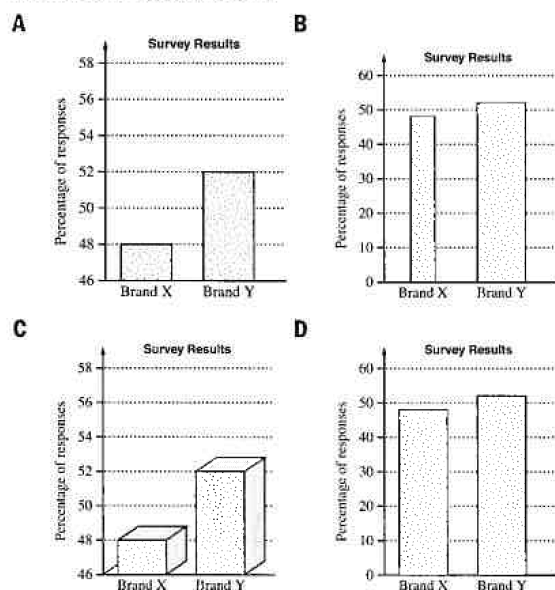
(Q4, 2015 HSC) **Easy**

- 6 A survey was conducted where people were asked which of two brands of smartphones they preferred. The results were:

- 48% preferred Brand X
- 52% preferred Brand Y.

A graph displaying the data is to be included in a magazine article. The editor of the magazine wishes to ensure that the graph is not misleading in any way.

Which graph should the editor choose to include in the article?



(1 mark)

(Q6, 2013 HSC) **Easy**

- 7 A high school has 100 students in each year group, Year 7 to Year 12. A survey is to be conducted to determine the average number of text messages sent per month by students at the school.

Which of the following would provide the most representative sample for this survey?

- A All Year 7 students
- B All Physics students in Years 11 and 12
- C 20 students chosen at random from each year group
- D 120 students chosen at random from the school roll

(1 mark)

(Q8, 2013 HSC) Easy

- 8 Handmade chocolates are checked for size and shape. Every 30th chocolate is sampled.

Which term best describes this type of sampling?

- A Census
- B Random
- C Stratified
- D Systematic

(1 mark)

(Q2, 2012 HSC) Easy

- 9 Greg needs to conduct a statistical inquiry into how much time people aged 18–25 years have spent accessing social media websites in the last two weeks. He has decided to survey a sample of students from his university.

The process of statistical inquiry includes the following steps, which are NOT in order.

- A Writing a report
- B Posing questions
- C Organising data
- D Analysing data and drawing conclusions
- E Collecting data
- F Summarising and displaying data

- i Using the letters A, B, C, D, E and F, list the steps in the most appropriate order for Greg to conduct his statistical inquiry. (2 marks) Easy
- ii Greg conducts his statistical inquiry. At which step in the process would he have drawn this graph?

Time spent accessing social media websites (in hours)



(1 mark) Easy

(Q26d, 2012 HSC)

- 10 A study on the mobile phone usage of NSW high school students is to be conducted. Data is to be gathered using a questionnaire.

The questionnaire begins with the three questions shown.

Q1: Do you own a mobile phone?
Yes ☐ No ☐

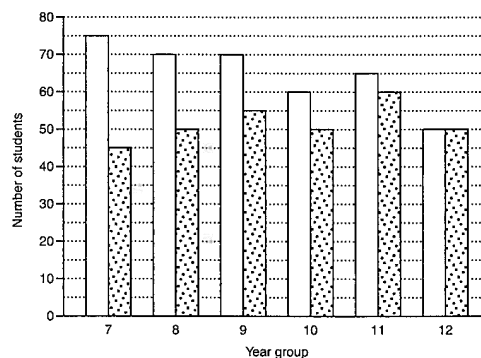
Q2: Which phone company do you use?
.....

Q3: Do you use pre-paid or a plan?
Pre-paid ☐ Plan ☐

- i Classify the type of data that will be collected in Q2 of the questionnaire. (1 mark) Easy
- ii Write a suitable question for this questionnaire that would provide quantitative data. (1 mark) Easy
- iii An initial study is to be conducted using a stratified sample. Describe a method that could be used to obtain a representative stratified sample. (1 mark) Medium

(Q25a, 2011 HSC)

- 11 The graph below displays data collected at a school on the number of students in each Year group, who own a mobile phone.



- ☐ Number of students in Year group
- ☒ Number of students who own a mobile phone

- i Which Year group has the highest percentage of students with mobile phones? (1 mark) Easy
- ii Two students are chosen at random, one from Year 9 and one from Year 10. Which student is more likely to own a mobile phone? Justify your answer with suitable calculations. (2 marks) Easy
- iii Identify a trend in the data shown in the graph. (1 mark) Easy

(Q25b, 2011 HSC)

- 12 The eye colours of a sample of children were recorded.

When analysing this data, which of the following could be found?

- A Mean
- B Median
- C Mode
- D Range

(1 mark)

(Q3, 2009 HSC) Easy

Classifying Data

- 13 Jamie wants to know how many songs were downloaded legally from the internet in the last 12 months by people aged 18–25 years. He has decided to conduct a statistical inquiry.

After he collects the data, which of the following shows the best order for the steps he should take with the data to complete his inquiry?

- A Display, organise, conclude, analyse
- B Organise, display, conclude, analyse
- C Display, organise, analyse, conclude
- D Organise, display, analyse, conclude (1 mark)

(Q5, 2009 HSC)

Easy

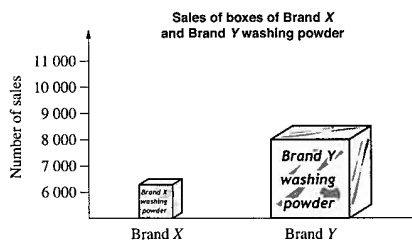
- 14 The Australian Bureau of Statistics provides the NSW government with data on the age of residents living in different areas across the state. After analysing this data, the government makes decisions relating to the provision of services or facilities.

Give an example of a possible decision the government might make and describe how the data might justify this decision. (2 marks)

(Q24c, 2009 HSC)

Easy

- 15 List TWO ways in which this graph is misleading.



(2 marks)

(Q24a, 2006 HSC)

Easy

- 1 Cumulative percentage for train or bus delay = 92%
Previous cumulative percentage = 86%
So percentage for train or bus delay = $(92 - 86)\%$
= 6%

Answer A

- 2 The responses are quantitative because they are numbers. The data is discrete because the responses are whole numbers.

Answer C

- 3 Testing every 50th item is systematic sampling.

Answer C

- 4 Blue, green, yellow
[Data in types that cannot be ordered]

Answer A

- 5 Categorical, ordinal data
[Data divided into groups which can be ranked.]

Answer A

- 6 The graph that should be included is D.
[The graph in option A doesn't begin at 0 on the vertical axis. The columns in graph B are not the same width and graph C uses 3D columns.]

Answer D

- 7 The most representative sample would have equal numbers of students from each year group. Choosing 20 students at random from each year group will provide the most representative sample.

Answer C

- 8 It is an example of systematic sampling.

Answer D

- 9 i B, E, C, F, D, A ✓✓ (2 marks)
ii A graph is a data display. This would be done in step F. (1 mark)

- 10 i The type of data will be categorical nominal.

(1 mark)

- ii (Any question that requires an answer that involves counting. Some possible answers are:)

'About how many phone calls do you make each day?'

'How many text messages did you send last week?'

(1 mark)

- iii (Any study that surveys students in proportion to the numbers in a particular group.)

A study might determine the number of boys and girls in each year level in urban, regional and remote areas and then select 1% of each of these groups to take part in a survey.

(1 mark)

- 11 i Year 12 has the highest percentage of students with mobile phones (100%).

(1 mark)

- ii In Year 9 there are 70 students and 55 have a mobile phone.

Percentage with mobile

$$= \frac{55}{70} \times 100\%$$

$$= 78\frac{4}{7}\%$$

✓

In Year 10 there are 60 students and 50 have a mobile phone.

Percentage with mobile

$$= \frac{50}{60} \times 100\%$$

$$= 83\frac{1}{3}\%$$

✓

So a Year 10 student is more likely to have a mobile phone.

(2 marks)

- iii (One possible answer:)

As the year level increases so does the percentage of students with mobile phones.

(1 mark)

- 12 Only the mode can be found for categorical data.

Answer C

- 13 Answer D

- 14 For example, data might show that in a particular area, there is a large number of young families, that is, a large number of pre-school age children. The government might then decide to provide more childcare services in that area. ✓✓

(2 marks)

- 15 1 By showing the sales of Brands X and Y as cubes (volume), the actual difference between the two brands has been exaggerated. ✓

- 2 The vertical axis representing sales does not start at zero. If it did, Brand X would not appear that much less than Brand Y. ✓

(2 marks)